IoT Traffic Monitoring and Alerts

CPSC 4240 – Fall 2026

Group Members: Emma Kropf, Anabel Thompson, Brayden Bennet, Sam DuBose

Project Overview and Objectives

Our objective is to build an open-source tool that processes IoT (Internet of Things) network data and alerts users to abnormal connections, unsafe protocol usage, and unusual bursts of requests.

Implementation Plan

We plan to use Wokwi to simulate network activity for an IoT device, download the pcap file, and programmatically analyze the traffic for red flags while also reporting overall diagnostics.

Systems and Live Demonstration

The main system we're interfacing with is the simulated IoT device. The in-class demonstration will include us collecting data from two devices, one behaving normally and one that simulates suspicious activity, to showcase functionality.

Why this Project is Useful

This project is useful because it helps users identify unusual IoT device communications and possibly insecure network activity. IoT devices are becoming more common, but most don't have strong security and perform unsafe activities. Our project will provide an easy way for people to add some level of protection to their devices.